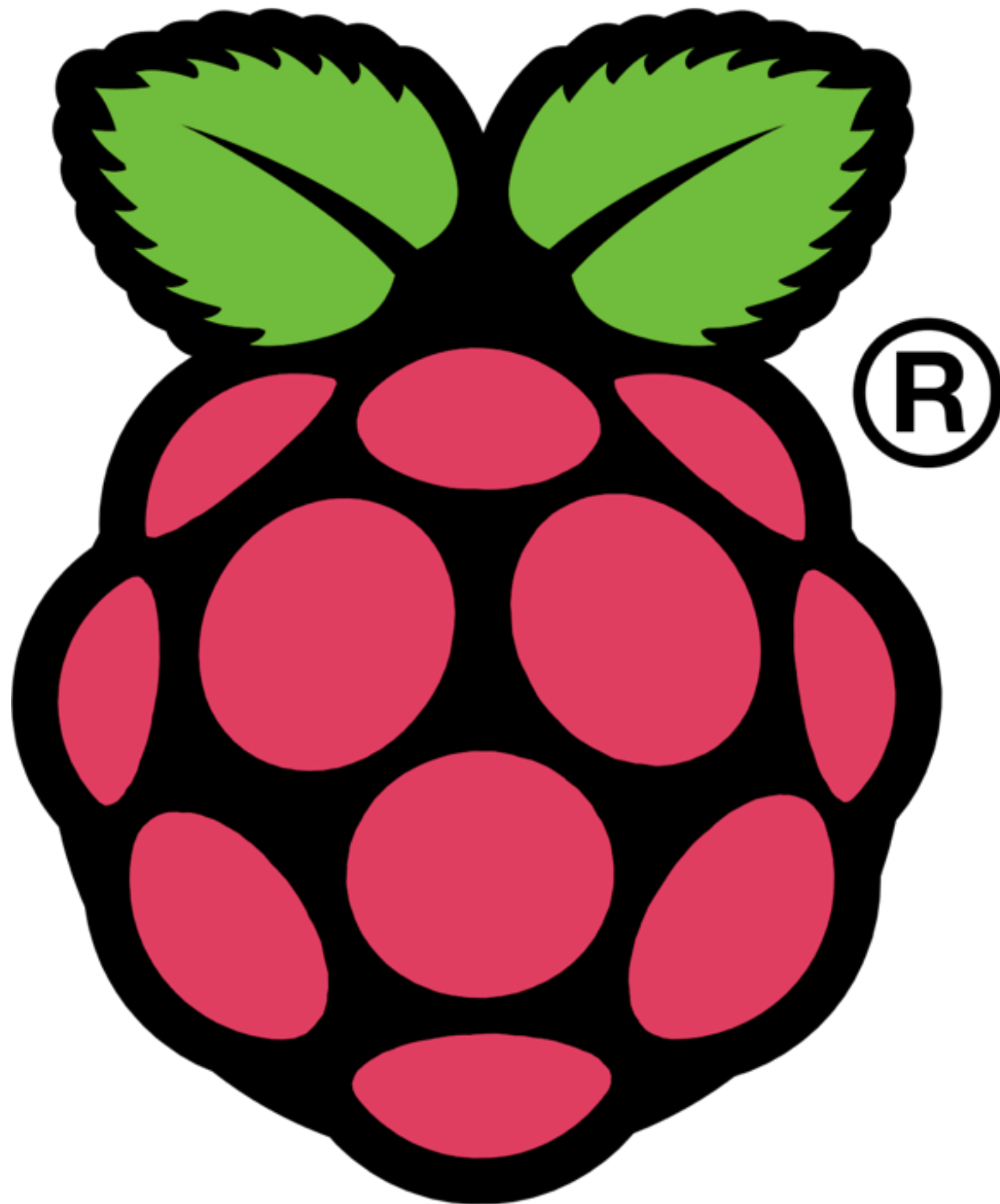


The Programming Language as a Musical Instrument

Sam Aaron







Clojure Training
&

Consultancy

<http://lambdanext.eu>



<http://overtone.github.io>



@overtone

“The laser beams sliced through the wafts of smoke as the subwoofer pumped bass deep into the bodies of the crowd...”

“The atmosphere was ripe with a heady mix of synths and dancing. However something wasn’t quite right in this nightclub...”

“Projected in bright colours above the DJ booth was futuristic text: moving, dancing, flashing...”

“This wasn’t fancy visuals, it was merely a projection of a terminal containing Emacs...”

*“The occupants of the DJ booth weren’t spinning disks,
they were writing, editing, and evaluating code...”*

“This was a Meta-eX gig. The code was their musical interface and they were playing it live.”

Making Music with Clojure

Live Coding is all the Rage at Raves

by Sam Aaron

Admit it: your real mission in life is to modify code in real time and have it projected on screens in a nightclub. Welcome to the wild world of Livecoding.



The laser beams sliced through the wafts of smoke as the subwoofer pumped bass deep into the bodies of the crowd. The atmosphere was ripe with a heady mix of synths and dancing. However something wasn't quite right in this nightclub. Projected in bright colors above the DJ booth was futuristic text: moving, dancing, flashing. This wasn't fancy visuals, it was merely a projection of a terminal containing Emacs. The occupants of the DJ booth weren't spinning disks, they were writing, editing, and evaluating code. **This was a Meta-eX^[U1] gig. The code was their musical interface and they were playing it live.**



And this wasn't a scene from a cheesy sci-fi film. Coding music like this is a growing trend and is often described as [Live Coding](#) ^[U2]. One of the recent directions this approach to music-making has taken is the [Algorave](#) ^[U3] — events where artists code music for people to dance to.

However, you don't need to be in a nightclub to Live Code music — you can

Meta-ex

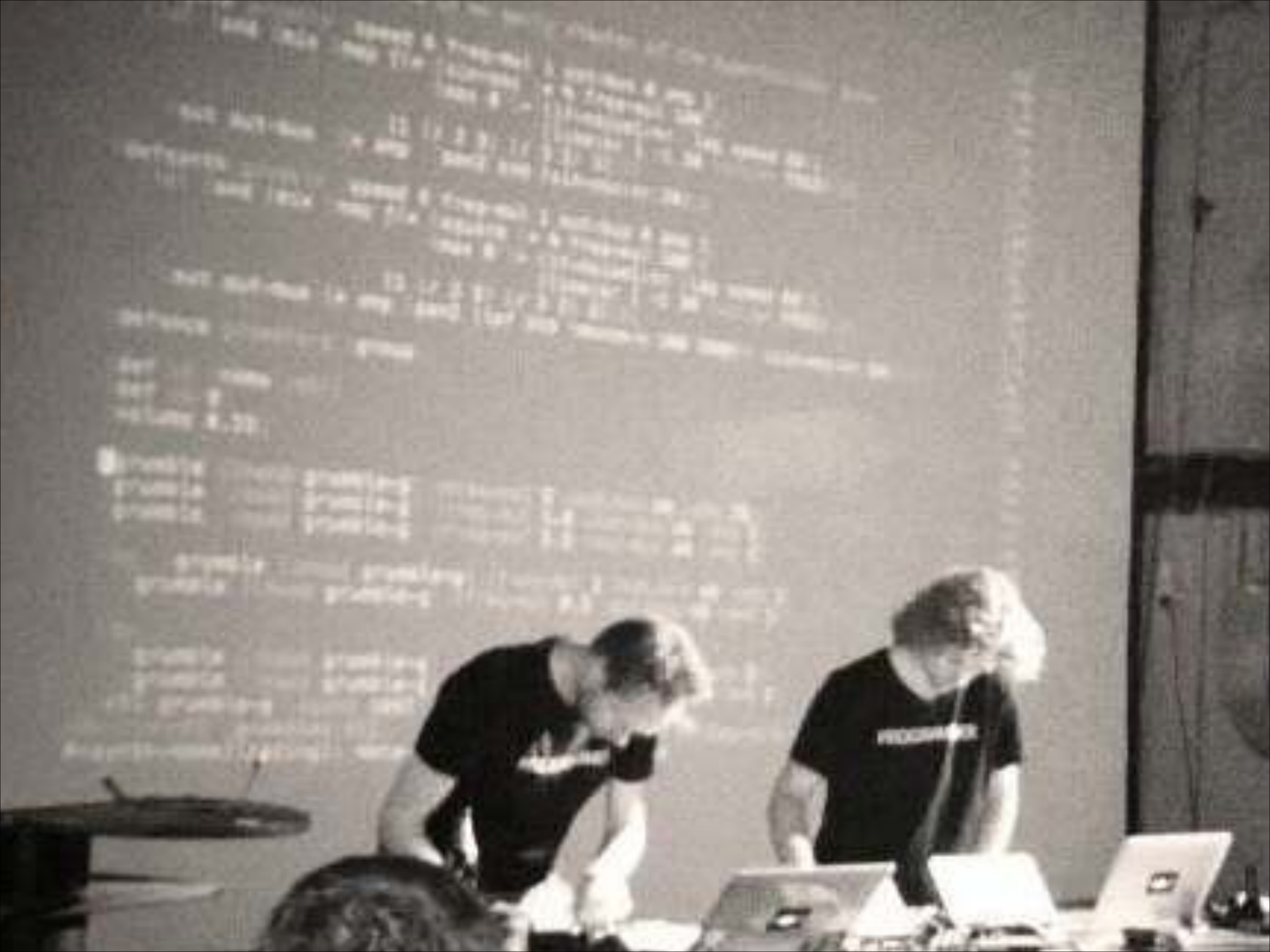
line











What is Programming?



What can programming
be?



How far dare we push
programming into new
territory?



Thought Experiment:

What if programming
was not
engineering?



Programming as Expression









Better Science Through Art

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Abstract

How do artists and scientists work? The same.

Categories and Subject Descriptors A.0 [General]

General Terms Design

Keywords Design, art, science, software engineering

~~~~~  
Prefrontal lightningbolt too lazy to chew the sphinx's loudest eyelash  
Not even if it shushes you with a mast of sneers  
Down which grateful bankvault-doors scamper  
Because of a doublejointedness that glows in the dark  
Like a soliloquy of walnuts  
Numbed by beaks of headless measuringtape  
So the lubriciousness can tower in peace  
Like a buzzsaw trapped in a perfumery of shrugs  
Lemon  
Or lime  
Only a maze can remember your hair of buttered blowguns

~~~~~  
From *Nights of Naomi* by Bill Knott [1]

~~~~~  
Art is strange. Art cannot be understood. The poet Robert Browning is reported to have said of a passage he wrote:



This is a sort of contemporary science which is more easily understood than the nonsensical poem.

Don't you think? Ah, but with training this is quite easily understood. The poem, however—obviously no amount of training, teaching, or learning will bring you to understand it.

Some other examples: a visual representation of a source code svn log shows, clearly, the history of the program, whereas the Jackson Pollock painting is clearly less understandable

Programming is a way of  
thinking



*“A programming language is a tool that has profound influence on our thinking habits.”*

Edsger Dijkstra



*"Programmers spend so much of their time  
in their own heads that trying to look at  
the world from someone else's viewpoint is  
a big shift"*

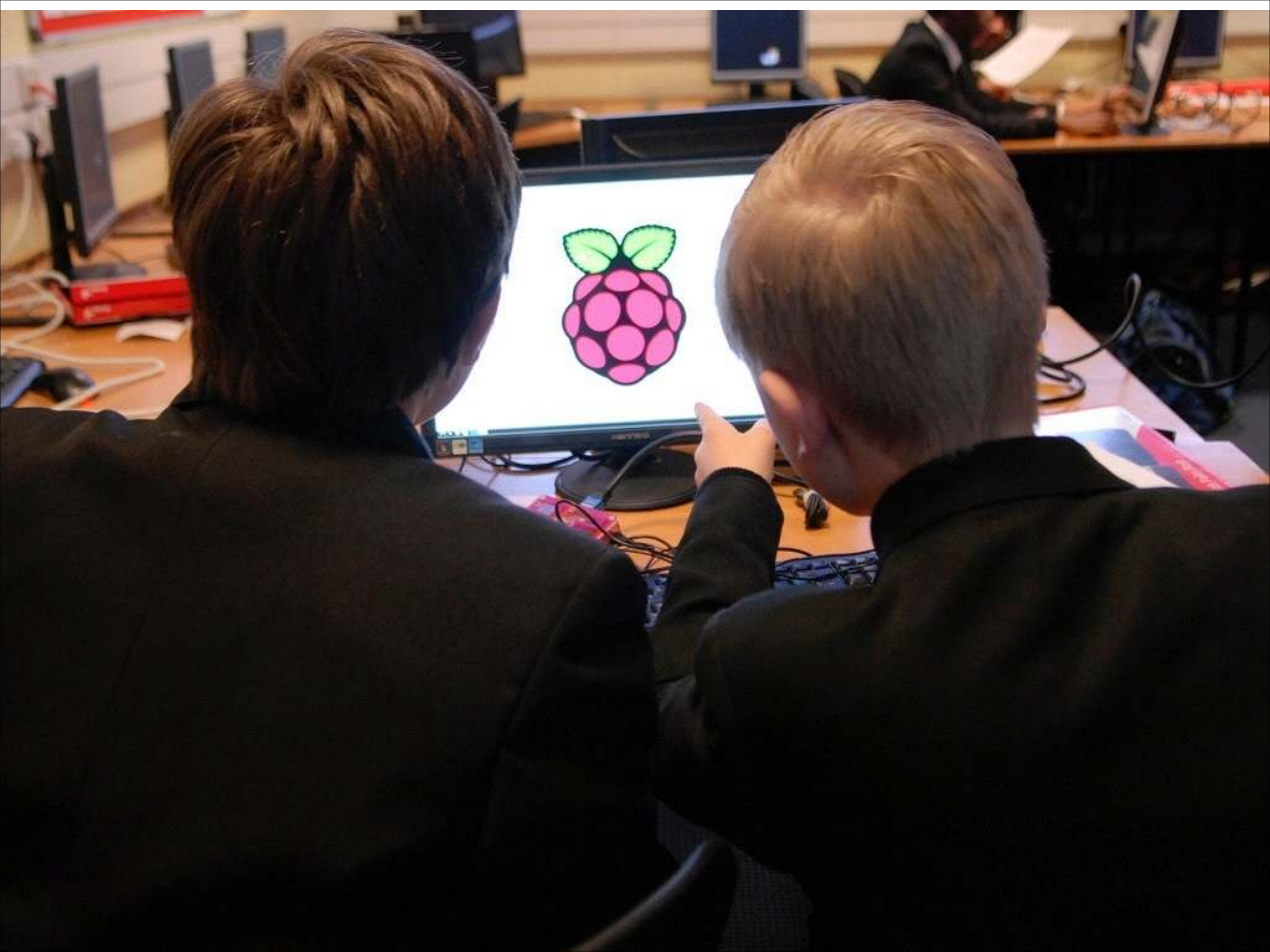
Kent Beck



*“A good programmer in these times does not just write programs, a good programmer builds a working vocabulary, in other words, a good programmer does language design though not from scratch but building from a frame of a base language.”*

Guy Steele

# Programming as a learning tool





## Making Computer Science Audible

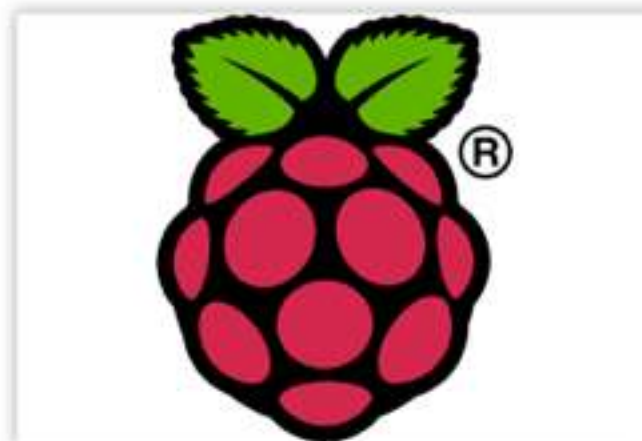
Sonic Pi is an open source programming environment designed to explore and teach new *programming concepts* through the process of creating new sounds. Sonic Pi comes with an associated **scheme of work** which emphasises the importance of *creativity* in the learning process and gives users *control* over what they want to do.

Make the music you want and have fun learning programming while you play.



### Classroom Ready

Sonic Pi comes with an associated **scheme of work** targetted for KS3 introductory Computer Science. This has been developed in harmony with the new curriculum proposed by the **CAS** working group.



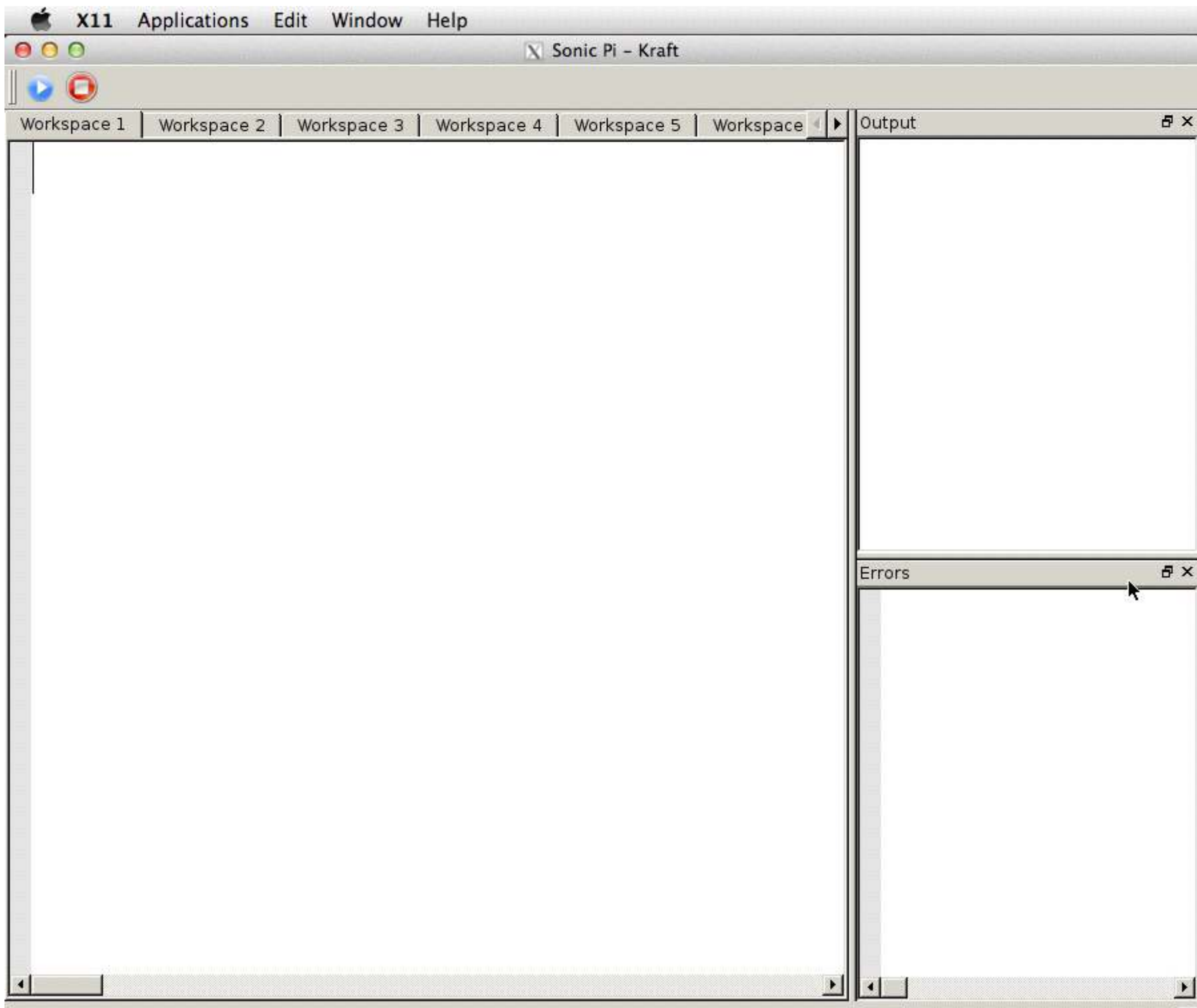
### Designed for Raspberry Pi™

Sonic Pi has been designed from the ground up to work with the **Raspberry Pi™**. All you need to convert your RPi into a music environment is some headphones or a speaker.



### Simple and Fun to Program

Sonic Pi has been designed to be as simple as possible to get started coding sounds and rhythms. Simply type your program, press play and hear results.



# Develop New Schemes of Work



Thank you for helping us  
make our wireless  
computers interesting  
and almost  
real life  
Ruth KB.

Thank you  
for helping  
us out XD  
Jordan  
Borden

Thanks for  
your help  
Mimpeax

Thanks  
for teaching us  
how to set up a  
computer & up a raspberry  
PI! Thanks  
Living

Thank you  
for teaching us  
from the Boss  
Alwore (AKA Karen).

Thanks  
Alfie ☺

THANKS  
from Beth...

Thanks ☺

Thanks  
for your  
help  
not  
wrong!

Roses are  
red, violets  
are blue...  
you made us  
pretty cool. ☺  
Charlotte N

Thank you for your help  
Michael Wilson

Hi,  
thanks for teaching  
us about the Raspberry  
Pi!  
Manah.)

Thank  
you  
Connor  
widmore

Thank  
you  
Victoria ☺

Thank  
you!  
Coby!

Thank  
you for  
helping us  
from  
Victoria ☺

TO  
Sam  
Thank you  
FROM

thank you  
-Carly ☺

Thanks  
for teaching  
me in a fun  
way!

Thank  
you  
for teaching  
us  
from  
Micaela  
McCann ☺

Thank you  
for teaching us  
that we need to have  
fun! ☺

Thank  
you  
from  
Charlotte

THANKS!  
from Alex M  
☺☺☺

Thank you  
from  
Charlotte N

Thank you  
for helping  
us with  
the Raspberry  
Pi ☺  
Gaby ☺

Thank you for the  
love and support  
☺

Thank  
you  
☺

Thank you  
for helping us  
with our  
Raspberry Pi's  
Lacey  
Brown

Thank you  
from  
Charlotte

Thank you for helping us  
from  
Charlotte

Thank you  
for helping  
Gaby ☺

Thank you  
from  
Charlotte

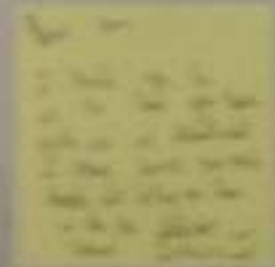
THANKS  
FOR  
YOU  
Time

Thank you  
from  
Charlotte

Thank you for helping us  
with our Raspberry Pi's  
Charlotte

Robert Clack School

8 Kilby + 8 Berners



Thank you  
for helping us  
out summer  
cha

Thank you for  
making dull lifeless  
computers interesting  
and almost  
reality

Ruth A.B.

x-next





# Programming as a form of dialogue

# Programming as an experience

Liveness

Feedback

# Exploration

Fail Early

Failure is Positive

Failure to perceive failure  
is failure

Conceptual efficiency is  
as at least as important  
as computational  
efficiency



*Thank you*

@samaaron

<http://sam.aaron.name>